Academic Reflective Journal

1. Introduction

The lab focused on Exploratory Data Analysis (EDA), an important stage in the data science process. EDA involves examining a dataset using statistics and visualizations to understand its structure, identify patterns, and recognize potential data-quality issues before building a machine learning model. The lab used the Titanic dataset to explore factors associated with passenger survival.

2. Description of the Experience

The lab began by introducing EDA and explaining the importance of understanding data before applying machine learning techniques. Using Python and Pandas, I examined the Titanic dataset with functions such as head() and info(). This allowed me to understand the variables, data types, and missing values in the dataset. Missing information was identified in the Age and Cabin columns.

I then used descriptive statistics with the describe() function. This provided information such as the mean, standard deviation, minimum, and maximum values. The analysis showed that the average passenger age was approximately 29.7 years and that the overall survival rate was approximately 38.4%.

The lab also introduced visual EDA using Matplotlib and Seaborn. Countplots were used to examine survival, passenger class, and gender. These visualizations showed that more passengers died than survived and that survival was strongly associated with passenger class and gender. The lab also used histograms to examine the age distribution of passengers according to survival status.

3. Personal Reflection

At the beginning of the lab, I viewed EDA mainly as calculating statistics and looking at a dataset. However, the activities helped me understand that EDA is a broader process of asking questions and using different techniques to discover patterns in data.

I found the visualizations especially useful because they made relationships easier to understand. The lab helped me understand the importance of connecting technical analysis with interpretation. Rather than simply producing a graph, I need to consider what it shows and why

the pattern may be important. This is particularly relevant when preparing data for machine learning because EDA can help identify variables that may be useful predictors.

The lab also connected with the theoretical concept of data preprocessing. Identifying missing values in the dataset demonstrates why data quality must be examined before a model is developed. I learned that incomplete or problematic data can affect later analysis and therefore needs to be considered during the early stages of a project.

One important critical-thinking lesson was that relationships observed through EDA should not automatically be interpreted as causal relationships. For example, although passenger class was associated with survival, the analysis alone cannot establish that passenger class directly caused survival. Other factors may also have influenced the results.

4. Improvements and Learning

The lab improved my technical skills in Python, particularly my ability to use Pandas for data inspection and descriptive statistics, and Seaborn and Matplotlib for visualization. I also became more comfortable interpreting graphs and using them to identify patterns.

A major area of personal growth was learning to approach a dataset systematically. In future projects, I will first examine the structure and quality of the data before beginning modeling. I will check for missing values, investigate numerical and categorical variables, and use appropriate visualizations to understand relationships.

I also recognize that I need more practice performing independent analysis. The lab included activities involving port of embarkation and fare, which provide opportunities to investigate additional relationships. Completing similar independent experiments in future work will help me become more confident in choosing appropriate visualizations and interpreting results without relying entirely on examples.

5. Conclusion

The EDA lab provided valuable practical experience in understanding and exploring data before machine learning. I learned how to inspect a dataset, identify missing values, calculate descriptive statistics, and use visualizations to investigate patterns. The most important lesson I learned was that effective data analysis requires both numerical and visual investigation. The Titanic dataset demonstrated how variables such as passenger class and gender can reveal important patterns in survival.